Global Facilities – Design & Construction - Mechanical – Water Based Fire Protection System

ECN Facility:	GLOBAL FACILITIES	ECN Area:	ENGINEERING
Owner(s)	ENGINEERING	Target Audience:	All Facilities
Document Type:	Standard	Document ID (pdf):	
Version:	1		A3YRXSD74VDV-57553043-397
Current ECN:	101074914	Last Modified:	11/10/2020

ECN History **Submit a Change Request**

1.0 Purpose

1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering, and Construction of new Water Based Fire Protection systems as part of new Greenfield Construction or major remodel projects.

1.2 The standard is applicable for Water Based Fire Protection systems in new construction or major remodel projects or buildings.

1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

2.1 This document covers new basebuild and major remodel Fire Protection System installations only. This document does NOT cover existing Water-Based Fire Protection systems nor does it require retrofit of existing systems to meet this Standard.

2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
AHJ	Authority Having Jurisdiction
AMHS	Automated Material Handling System Area per Sprinkler Head design term for the floor space that a sprinkler head is designed to protect.
Demand Area	the hydraulically most remote area of the water-based fire protection system. Measured in square feet or square meters.
Demand Density	Water discharge in gpm/min or mm/min
FAB	Semiconductor fabrication facility
FATC	Fire Alarm Terminal Cabinet
FM or FMG	Factory Mutual Global Insurance Company
FOUP	Front Opening Unified/Universal Pod
HVM	High Volume Manufacturing facility
HPM	Hazardous Production Materials
IBC	International Building Code
ICC	International Code Council
IFC	International Fire Code
k-factor	a performance metric for the flow of water from a sprinkler head. It is equivalent to the Cv factor for hydraulic pressure drop calculations.
LOTO	Lock-Out, Tag-Out
NFPA	National Fire Protection Association
PAFS	Pre-Action Fire Sprinkler system
QR	Quick Response sprinkler head
SR	Standard Response sprinkler head

UL	Underwriters' Laboratory

4.0 Roles and Responsibilities

Role	Responsibilities
GFTT Lead / Risk Management Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead/Risk Management Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

The Fire Protection systems covered within this document use water as the primary method to control the spread of a fire within a space or a building. There are also Specialty Fire Protection/Suppression systems that use an inerting/ chemical agent or a combination of water and additives. Refer to **Table 1** for an overview of the applications, locations, and systems covered in this Standard. Guidance is also provided for those system types that are covered in separate Standards. In some locations and/or occupancies, more than one system type may be an acceptable option. For example, an Electrical Room can be protected by a Wet Pipe system, a Pre-Action system, or a Chemical or Gaseous Fire Suppression system. Water Based Fire Protection systems will generally provide the best form of fire protection on the most cost-effective basis.

5.1 **Wet Pipe Systems** use distribution piping that normally contains water. When a localized fire, detonation air wave, or heat source contains sufficient heat energy, a sprinkler head(s) will fuse/open. Water will then flow thru the open head(s) and create a developed flow profile to mitigate fire spread from the event. As described, **the Wet Pipe system shall be the default** system design for Micron sites.

5.2 **Dry Systems** are filled with pressurized gas (CDA or nitrogen). The pressurized gas will hold the dry-type "flapper" valve shut at the riser room. During an event, the heat will cause a normally closed sprinkler head to fuse (open). The open head will allow the pressurized gas to escape. As the gas escapes thru the open head, the pressure is relieved from the flapper valve on the dry type valve, allowing water to fill the pipes. The water will only flow at the open head locations.

5.3 **Pre-Action Fire Sprinkler (PAFS)** systems use a distribution piping arrangement that does not normally contain water. The system piping is typically filled with pressurized gas (CDA or nitrogen) at a relatively low pressure to monitor the integrity of the sprinkler system piping. Activation of the "deluge valve" on the system riser is generally through a fire detection device (e.g. heat or smoke detector). Upon activation, the automatic "deluge" valve will open and charge the system with water, but water may not flow until a sprinkler head fuses. This type of system is generally reserved for locations that are susceptible to water damage, such as, IS machine rooms, electrical rooms, etc.

5.4 **Deluge Sprinkler Systems** are very similar to a PAFS systems except that they employ "open" sprinkler heads. An activation device (fire detector or pilot line comprised of piping, sprinkler heads, and compressed air) will trigger the deluge valve to open. Upon activation, water is admitted to the piping and is dispensed through **<u>all</u>** sprinkler heads. Typical applications of a Deluge system include bulk silane pads, large outdoor transformers, and cooling towers.

Table 1 – Fire Suppression System Types

Fire Suppression System Type	Typical Spaces Covered by System Type
Wet Type Systems	
Wet Pipe Systems	• FAB Cleanroom, Clean Subfab, Utility Subfab • FAB Plenum or Interstitial levels where required. • Chemical Rooms (Acids, Caustics, etc.) • Gas Rooms • Utility Areas (Central Plant Mechanical Rooms, Water/Wastewater Plant) • Standard Building Spaces (i.e. Offices, Corridors, Lobbies, Toilet Rooms, etc.) • Fire protection inside combustible (i.e., poly vinyl chloride (PVC), polypropylene (PP), fiberglass reinforced plastic (FRP) fume exhaust ductwork • Preferred system for I/S machine (computer) rooms and electrical rooms
Dry Type Systems	

Dry System	• Cooling Towers
	• Areas subject to freezing conditions
	    ○ Docks
	    ○ Unheated warehouses
	    ○ Epoxy storage coolers

Pre-Action Type	• Is an acceptable alternative for electrical rooms (i.e. Substation, Transformer, Electrical), computer rooms/server rooms • Areas subject to or highly susceptible to water damage
Deluge Type	• Bulk Silane Pad • Bulk Ammonia Pad • Cooling Towers
Specialty Type Systems	
Specialty – Foam (*Separate BKM*)	• Solvent Rooms (i.e. Organic & Solvent storage or waste, IPA, PGMEA)
Specialty – Gas and Chemical Suppression (*Separate BKM*)	• Kitchens • Chemical Dispense Units (CDU's) • On-board production tools • Includes extinguishing/suppression agents such as CO2, Halon, Inergen, etc.

6.0 System Design Methods

The piping system must be designed to ensure sufficient pressure exists at any point within the system for adequate water flow and pressure to meet stated design requirements and suppress a fire. Piping configuration/ arrangement, pipe lengths, and pipe diameter/sizing are the most influential hydraulic design factors.

6.1 Distribution Piping Layout

6.1.1 Loop/Grid shall be the default configuration for wet pipe systems. It is the most hydraulically efficient, cost-effective design. Loop/grid shall not be used for Dry Type systems.

6.1.2 Tree/Branch piping arrangement is acceptable but not preferred. Tree/branch systems typically use larger piping to deliver the same hydraulic performance as a Loop/Grid system.

6.2 Distribution Piping Sizing

6.2.1 Pressure drop (as a function of flow rate) through the system is the primary design consideration for pipe sizing.

6.2.2 For the same flow rate, larger pipes exhibit lower pressure drop than smaller pipes. Installation of pipes larger than necessary consumes space and is not cost efficient.

6.2.3 With respect to **Section 6.1**, the loop/grid configuration minimizes the required pipe diameter sizes compared to the Tree/Branch arrangement, thereby minimizing material costs.

6.2.4 See **Section 8.6** for details on pressure drop and hydraulic calculation requirements.

7.0 System Components

All components within a Fire Sprinkler system must be approved for their intended use within the system. It is strongly recommended that all components are FM approved or equivalent. In some jurisdictions, such as Japan, FM approved devices are not readily available on a cost-effective basis. GFTT and/or Risk Management should be consulted to develop equivalent options to FM approved devices. The AHJ is the final approving entity for materials and components within a system.

Note: The availability of materials or manufacturers listed will vary from site to site. Local Construction Specifications shall take precedence over any manufacturers listed. The manufacturers listed below are only examples of possible suppliers. Manufacturers beyond those listed may be used at the discretion of the AHJ and the Design Engineer.

The terms and components listed below are specific to Water-Based Fire Protection systems.

7.1 Valves

7.1.1 Post Indicator Valve (PIV) is an underground valve, typically outside of the riser room for a building. The hardware of the PIV allows for surface manipulation of the underground valve. A PIV identifies the valve position of the primary isolation valve for the water source of the building.

a) Approved Manufacturers - Anvil International, Mueller Co., Tyco Fire & Building Products, or approved equivalent

7.1.2 Curb Box Valve is an underground valve used to isolate the primary water source for the building or through the piping distribution system. A curb box valve is like a PIV, except that it does not have an indicating post. A specialized valve wrench is necessary to reach below surface level and manipulate the valve position. A curb box valve is an effective means of isolation when a PIV cannot be used, such as, in a roadway.

a) Approved Manufacturers - Anvil International, Tyco Fire & Building Products, Victaulic, or approved equivalent

7.1.3 Position Indicating Butterfly Valve (IBV) can be used to isolate an entire zone at a riser room or isolation for a sub-system as an alternate to an OS&Y valve.

a) Approved Manufacturers - Anvil International, Tyco Fire & Building Products, Victaulic, or approved equivalent

7.1.4 Outside Stem & Yoke (OS&Y) is one type of valve used for isolation of a building or zone on a Fire Protection system. Typically, OS&Y valves are used on a sprinkler system riser.

a) Approved Manufacturers – Milwaukee Valve Co., NIBCO Inc., Victaulic, or approved equivalent

7.1.5 Ball valves are typically used on piping sizes 2” (50mm) and smaller. A ball valve can be used for isolation of a smaller sub-systems or for ancillary functions on a zone. Ancillary functions may include drains, Inspector’s test ports, etc.

a) Approved Manufacturers - Anvil International, Kennedy Valve, Tyco Fire & Building Products, or approved equivalent

7.1.6 Check valves allow flow in one direction only. They are typically installed downstream of a zone isolation valve in the Riser room. Check valves prevent flow across sprinkler zones/systems and protect the quality of the main water supply.

a) Approved Manufacturers – Reliable Automatic Sprinkler Co, Inc., Tyco Fire & Building Products, Victaulic, Viking, or approved equivalent

7.1.7 Monitored valves / tamper switches provide an electronic signal (usually routed back to a Fire Alarm or Life Safety Systems Control Panel) that indicates the status of the valve (i.e., open or closed). Monitoring hardware can be installed on a wide variety of valve listed above, with the exception of a curb box valve.

7.2 Water Flow Indicators (also known as Flow Switches)

7.2.1 Flow switches provide an electronic signal/alarm back to the FATC to identify when water flow is detected on a particular zone.

7.2.2 Alarm descriptions shall be unique to each zone.

7.2.3 Alarm descriptions shall allow responders to easily identify the area of impact.

7.2.4 Refer to the Global Design Standard for Instrumentation & Controls for approved flow switches.

7.2.5 Approved Manufacturers – Potter Electric Signal Company, System Sensor (Honeywell), or approved equivalent

7.3 Piping

7.3.1 All piping shall conform to ASTM A53/A53B Standards (JIS G3452).

7.3.2 Piping installed in Cleanroom airflow areas shall be either painted black steel or galvanized steel.

7.3.3 Piping installed in corrosive areas, such as Chemical rooms and Wastewater Treatment areas, shall be galvanized steel.

7.3.4 Piping installed in non-Cleanroom areas shall be black steel.

7.3.5 Refer to the project specific Construction Specifications for additional requirements on materials, joining methods, and approved manufacturers.

7.4 Fittings

7.4.1 For piping 2" and smaller, threaded fittings shall be used.

7.4.2 For piping 2-1/2" and larger, connections may be made via any of the following methods:

a) Approved rolled & grooved fittings
b) Flanges/bolts
c) Welding

7.4.3 Refer to the project specific Construction Specifications for additional requirements on materials, joining methods, and approved manufacturers.

7.5 Flexible Sprinkler Hoses (i.e. Flex Heads)

Flexible sprinkler hoses are typically used as the terminal connection between rigid piping and the final location of sprinkler heads within the ceiling grid or exhaust ductwork. They allow for flexibility of remodels and slight misalignments of piping and ceiling grid. Flex sprinkler hoses carry a higher pressure drop (i.e., higher K-factor **see 7.6.1**) than rigid pipe.

7.5.1 Flexible sprinkler hoses shall consist of stainless steel braided hose, with length 5 to 6 feet (1.5 to 1.8 meters).

7.5.2 Flexible sprinkler hoses shall be rated for the specific application, including but not limited to, space occupancy type, hazard classification, inside combustible fume exhaust ducts, etc.

7.5.3 Flexible sprinkler hoses shall be compatible with the ceiling grid type and mounting hardware.

7.5.4 Approved Manufacturers: Victaulic VicFlex Series AH2 (preferred), Flexhead Industries, Reliable RASCOFLEX, or approved equivalent.

7.6 Sprinkler Heads

Proper selection of sprinkler heads is of critical importance in any Water Based Fire Protection system. In all cases, the sprinkler heads selected must be approved for the application and occupancy.

7.6.1 k-factor is a numerical representation of the ability of a sprinkler head to deliver water. A higher k-factor indicates higher water flow at a given pressure. Certain occupancies require specific k-factor heads to mitigate an event.

7.6.2 Temperature Rating indicates the estimated temperature required to fuse/open a sprinkler head.

7.6.3 Response Type is either Quick Response (QR) or Standard Response (SR) for non-storage applications. A QR sprinkler head is the default for wet pipe systems. Early Suppression Fast Response (ESFR) sprinkler heads may work for certain storage locations.

7.6.4 The sprinkler head type for each designated space shall be specified per **Table 2**.

7.6.5 Protective guard covers shall be provided for all sprinkler heads located above critical areas including, but not limited to, FAB Cleanroom, Subfab, Interstitial Plenum, and FAB Auxiliary support rooms/areas, Utility Room areas, HPM Areas, Electrical Rooms, Server Rooms, etc.

7.6.6 Re-use of sprinkler heads is forbidden. Only new sprinkler heads may be installed in a system. Once removed from a system, a sprinkler head shall not be re-installed in any Micron sprinkler system.

7.6.7 Approved Manufacturers: Viking, Reliable, Tyco, or approved equivalent

7.7 High Point Vents

7.7.1 Oxygen is a primary contributor to corrosion within wet pipe systems. High point vents within a system or zone allows trapped air and non-condensable gases at the air-water interface to escape from a wet pipe system.

7.7.2 A typical high point vent will utilize two float assemblies plumbed in series with a leak detection device between the two floats. Contact GFTT and/or Risk Management for details associated with the high point vents.

7.7.3 A high point vent shall be installed at the highest elevation within each zone or sub-system.

7.7.4 Approved Manufacturers – Engineer Control Systems (ECS), Potter Electrical Signal, or approved equivalent

Example of a high point vent assembly

7.8 N_2 Inerting Stations

7.8.1 An N_2 inerting station will admit nitrogen gas into the sprinkler system piping to purge oxygen from the system. The nitrogen will act as a substitute for ambient air, which contains oxygen and accelerates the corrosion of piping in a wet pipe system.

7.8.2 An N_2 inerting station shall be installed and connected to each zone within the riser room.

7.8.3 Approved Manufacturers – Engineer Control Systems (ECS) or approved equivalent

8.0 Design Criteria

8.1 Occupancy Types and Demand Density

The performance requirements of any sprinkler system inside a building are dependent upon the occupancy type, which identifies the nature of the hazard within the space. An occupancy with a high hazard will require a higher water density than an occupancy of a lower hazard. See **Table 2** for typical occupancy types and demand density. For any occupancy type not covered, contact Risk Management for guidance.

8.2 Sprinklers - Area of Coverage and Head Spacing

8.2.1 Sprinkler heads are typically aligned to the ceiling grid, on either an 8’x12’ (2.44m by 3.66m) or a 10’x12’ (3.05m by 3.66m) spacing pattern.

8.2.2 See **Table 2** for standard/typical design requirements for nominal Design Demand Area and nominal Area per Sprinkler Head guidelines, as a function of Occupancy.

8.2.3 Provide the sprinkler heads at proper locations, spacing, and elevations as required to comply with all NPFA, local code, and AHJ requirements for proper sprinkler coverage.

8.2.4 Coordinate sprinkler head locations with all potential obstructions below, including but not limited to building structure, utilities, equipment, AMHS, manufacturing tools, etc., which may affect proper sprinkler coverage.

8.2.5 Sprinkler heads shall be located a minimum distance and elevation away from building structure and other obstructions per NFPA, local code, and AHJ requirements, to provide proper sprinkler coverage and prevent potential damage to the sprinkler heads.

8.3 Performance Specs

8.3.1 Applicable more restrictive local code requirements at each location supersede any requirements contained within this document. The performance requirements of any system will be dependent upon the type of occupancy.

8.3.2 In the absence of code direction, the fire sprinkler system shall be designed and installed per the Standard/Typical Design Requirements as outlined below in **Table 2**.

Table 2: Standard/Typical Design Requirements for Fire Sprinkler Systems

Occupancy	Demand Density, gpm/sqft (mm/m2)	Design Demand Area, sqft (m2)	Area per Sprinkler Head[1], sqft (m2)	Sprinkler Type
Cleanroom Areas (FAB, Subfab, Plenum2)	≥0.20 (≥ 8.1)	3,000 (279)	≤120 (≤ 11.14)	Quick Response, k ≥ 8.0 (k ≥ 80.6)
FAB Building Utility Areas (Mechanical	≥0.20 (≥ 8.1)	3,000 (139)	≤120 (≤ 11.14)	Quick Response, k ≥ 8.0 (k ≥ 80.6)

Occupancy	Demand Density, gpm/sqft (mm/m2)	Design Demand Area, sqft (m2)	Area per Sprinkler Head[1], sqft (m2)	Sprinkler Type
Rooms, Electrical Rooms, etc.)				
HPM Areas (Chem Rooms, Gas Rooms, etc.)	≥0.30 (≥ 12.2)	2,500 (232)	≤120 (≤ 11.14)	Quick Response, k ≥ 8.0 (k ≥ 80.6)
Utility Areas (Mechanical Rooms, etc.)	≥0.20 (≥ 8.1)	1,500 (139)	≤120 (≤ 11.14)	Quick Response, k ≥ 8.0 (k ≥ 80.6)
General Building Spaces (Office Areas)	≥0.10 (≥ 6.1)	1,500 (139)	≤225 (≤ 20.90)	Quick Response, k ≥ 5.6 (k ≥ 80.6)
Fume Exhaust Ducts[3]	Minimum discharge shall be 76 L/min (20 GPM) per sprinkler	From the five hydraulically most remote sprinklers.	Sprinklers shall be spaced a maximum of 6.1 m (20 ft) apart horizontally and 3.7 m (12 ft) apart vertically.	Wax coated, Quick Response, Double plastic bags k ≥ 5.6 (k ≥ 80.6)

[1] Refer to **Section 8.6** for specific guidelines regarding hydraulic calculations and area per sprinkler head.
[2] Sprinkler protection requirements may or may not be required in Cleanroom Plenum areas, based on local jurisdiction and code requirements
[3] Interior automatic sprinklers shall not be required where ducts are approved for use without internal automatic sprinklers. One example of an approved duct would be Teflon-lined stainless steel. Refer to the Global Standard for Process Exhaust Systems for guidance regarding installation of sprinklers in ducts.

8.4 Water Supply

8.4.1 The source of water for the sprinkler system may be from a municipality or an industrial water supply.

8.4.2 A dual lead-in to the riser room of a building from the primary water source is recommended. This requires a special valving arrangement.

8.4.3 Each site shall have a redundant water supply. Redundancy of water supply can be achieved by either of the following:

a) Secondary, independent feed from municipality
b) Gravity (or pumped) feed from retention tank/vault. The retention tank/vault shall be sized at a minimum to provide water supply to the largest demand area flow, for a minimum 20-minute duration, unless otherwise required by the AHJ or Local Code.

8.5 Water Supply Curve

The Water Supply Curve is a numerical and semi-log graphical representation of flow and pressure characteristics of the water distribution system.

8.5.1 Each site should conduct annual hydrant flow tests. The test results shall be recorded by considering the static pressure (zero flow) and residual pressure (actual fire flow) from the water supply.

8.5.2 If a fire pump is installed, a separate flow test shall be conducted annually. This test shall look for the pump performance relative to the OEM pump curve or a theoretical pump curve.

8.5.3 The volume of the tank for fire pumps shall contain at least 4 hours of flow to support the Demand area.

8.5.4 The water tank shall be properly sized to ensure adequate volume / capacity exists to meet worst-case flow for sprinklers and hydrants over the specified design duration.

8.6 Demand Area Hydraulic Calculations

The Demand Area is the hydraulically most remote area of a water-based fire sprinkler system. This theoretical demand area is used during the overall fire sprinkler system design. Depending on the piping arrangement of the system, the Demand Area can be identified using different methods.

8.6.1 All hydraulic calculations (flow vs. pressure loss) shall be performed by an approved hydraulic loss calculation program.

8.6.2 Ensure that any hose flow requirement is captured in the hydraulic calculations. A minimum hose flow demand of 250 gpm (950 lpm) is typical for semiconductor fabrication facilities; some more demanding applications, such as, solvent rooms and cooling towers, require up to 500 gpm (1900 lpm) hose flow allowance. Refer to your specific local code or AHJ for each location and application. Contact GFTT and/or Corporate Risk Management for further details.

8.6.3 The Water Supply Curve shall maintain a minimum of a 5% cushion above the Demand Area conditions, including hose flow requirements.

8.6.4 For hydraulic calculations, the designer shall use the maximum area per sprinkler head that is allowed in Code.

8.6.5 The maximum area per sprinkler head allowed by Code is usually greater than the nominal design area per sprinkler head. Hydraulic calculations will not use the design area per sprinkler head, nor the actual area per sprinkler head.

EXAMPLE: In the United States, the IFC limits a sprinkler head within a clean room to a maximum of 130 ft^2 of coverage area. The ceiling is arranged in a 2′ x 4′ grid array. The designer chooses to set a nominal design area per sprinkler head in an 8′ x 12′ array to align with the ceiling grid. The nominal design area is now 8′ x 12′ = 96 ft^2 per sprinkler head. The hydraulic calculations will be run using the 130 ft^2 per sprinkler head.

8.6.6 By using the maximum area per sprinkler head allowed, this creates a cushion that will allow for flexibility in the case of future construction efforts, i.e. wall moves, AMHS track construction, etc.

8.7 Fire Pumps

8.7.1 Fire pumps are intended to ensure uninterrupted service of the Fire Protection system, should the primary water supply become unstable or fail.

8.7.2 There are two basic type of fire pumps; a booster pump and a fire pump.

a) A booster pump takes suction from a water supply, such as an industrial or municipal water supply and boosts its pressure. The booster pump can provide a cost-effective means to increase system flow and pressure providing the industrial / municipal water system has adequate flow and pressure to support the booster pump.
b) A fire pump takes suction from a water source such as an above ground tank or below grade tank / vault.

8.7.3 When deemed necessary, fire pumps shall be sized for the single largest Demand Area across all buildings within the entire site.

8.7.4 The installation of a dedicated fire pump for each building is discouraged.

8.7.5 Installation arrangement of fire pumps shall include redundancy. Two fire pumps and/or booster pumps shall be provided for backup and redundancy, and shall consist of the following:

a) Electric powered pump (primary) - The Electric powered pump shall be connected to emergency power. Connection to an Uninterrupted Power source (such as UPS or DC Bank) is preferred.
b) Diesel powered pump (redundant/secondary)

8.7.6 Approved Manufacturers – Armstrong, Bell & Gossett, Peerless, or approved equivalent

8.8 Sprinkler Water Containment

8.8.1 Sprinkler Water Containment shall be provided to avoid unintentional discharge or release of hazardous chemicals and fire sprinkler water. Containment is primarily required for solvent rooms containing flammable and combustible liquids.

8.8.2 The designer shall consider and account for containment of sprinkler water and the worst-case discharge of Hazardous Production materials, in the case of an activation of the Fire Protection system.

8.8.3 The containment volume shall account for a minimum of 20 minutes sprinkler discharge duration of the Demand flow.

8.8.4 For rooms with chemical tanks or drums, in addition to the volume required by 20 minutes (minimum) of Demand Area flow, the containment volume shall be a function of the tanks/drums.

a) All tanks/drums are metal. The volume of the largest single chemical tank within the room shall be added to the containment volume.
b) Tanks/drums are plastic. The total volume of ALL plastic containers shall be added to the containment volume.

9.0 System Applications

9.1 Cleanroom Spaces

9.1.1 Cleanroom Spaces, including FAB level and Subfab Level shall be provided with Wet Pipe Sprinkler systems.

9.1.2 AMHS systems, including support structures and FOUP's can present significant obstructions to proper sprinkler pattern discharge. Coordinate the sprinkler head locations with the AMHS system to ensure proper sprinkler coverage.

9.1.3 If the Interstitial Plenum level above a Cleanroom requires fire sprinkler protection per the local code or AHJ requirements, it shall be provided with a Wet Pipe Sprinkler system.

9.1.4 Sprinkler heads shall be connected to the Cleanroom ceiling grid with flexible hose assembly.

9.1.5 In certain circumstances, a sprinkler system below a raised floor structure in a Cleanroom may be omitted. This is dependent on local code and AHJ requirements.

9.2 General Building Spaces

9.2.1 General Building Spaces (such as Offices, Conference Rooms, Lobby Areas, Corridors, etc.) shall be provided with Wet Pipe Sprinkler system.

9.3 Electrical Rooms, Computer Rooms, Server Rooms

9.3.1 Electrical Rooms, Computer Rooms, Server Rooms, etc. shall be provided with a Wet Pipe Sprinkler System or a PAFS.

9.4 Hazardous Production Material Rooms (Chem/Gas Rooms)

9.4.1 HPM Rooms for Acid and Caustic Chemicals and Gases shall be provided with Wet Pipe Sprinkler system.

9.4.2 HPM Rooms for Solvent Chemicals and Solvent Waste shall be provided with Specialty Foam Fire Suppression system. The requirements for these systems shall be covered in a separate System Standard document.

9.5 Exhaust Ductwork

9.5.1 Refer to the Global Standard for Process Exhaust Systems and **Table 2** in this document for additional sprinkler requirements regarding Exhaust Ductwork.

9.5.2 Sprinkler heads shall be provided inside the exhaust ductwork, for ductwork constructed of combustible materials, sizes 10 inches (254 mm) and greater in diameter, or with cross section area equal or greater to 80 inches^2 (516 cm^2). This is per NFPA 318 Section 4.1.2.6 and FM Standard 7-78.

- Internal fire sprinklers are not required for FM Approved ductwork.

9.5.3 Sprinkler heads shall be provided on the inlet and outlet connection of each Exhaust Scrubbers. The sprinkler head can be located on either the inlet and outlet ductwork, or on the scrubber body inlet and outlet connections.

9.5.4 If required by local code, provide internal sprinkler heads inside ductwork on both sides of fire rated partitions.

9.5.5 Sprinkler heads installed inside Acid and Ammonia exhaust ductwork shall be corrosion resistant and/or protected from corrosion. Allowable types include FM approved wax, lead, or wax-over-lead coated sprinkler heads, or plastic-coated sprinkler heads (coated with epoxy, TFE resin, etc.). Additional double layer polypropylene or polyethylene bag may be used to contain the sprinkler head.

9.5.6 Sprinkler heads shall be connected to the sprinkler branch piping with flexible hose assembly.

10.0 System Installation

10.1 Piping Arrangement

10.1.1 Connection Methods and fittings — refer to **Section 7.4**

10.1.2 Piping take-offs shall always occur at or above the horizontal plane. Connections below the horizontal plane are prone to scale accumulation. During an event, this scale may flow to the open heads and will prevent proper sprinkler head performance.

10.1.3 Ideally, take-offs should come from the top of the pipe and include the use of return bends.

10.1.4 Where possible, horizontal mains should be sloped upwards as they leave the riser room. This will allow for more complete draining and filling of the system.

10.1.5 The combination of sloping, proper placement of High Point Vents, and the installation of N2 Inerting Stations will extend the life of the system by minimizing corrosion potential and greatly improve the Operation and Maintenance of the overall systems.

10.2 Sub-systems

10.2.1 Sub-systems are installed where a portion of the overall system will experience numerous modifications to support a specific need.

10.2.2 The use of sub-systems is especially useful during fit-up of tools in a newly constructed building. It allows for a small segment of the system to be modified without removing Fire Suppression from an entire sub-fab, for example.

10.2.3 Each sub-system shall have its own dedicated, monitored isolation valve. Examples of possible sub-system applications include, but are not limited to, the following:

a) Gas Cabinet or Liquid Source Cabinet areas in a Subfab
b) Gas bunkers or chemical storage rooms
c) Flammable exhaust duct coverage

10.3 Hangers and Seismic Bracing

10.3.1 Piping for Fire Protection systems shall be supported independently of any other systems or utilities.

10.3.2 Hangers and seismic bracing shall be designed and installed using an approved computer design program.

10.3.3 Seismic design parameters shall be based on FM Global Data Sheet 2-8 "Earthquake Protection for Water Based Fire systems". Properly identifying the seismic zone is critical for proper design.

10.3.4 For dry pipe systems, significantly higher forces (water hammer) will be generated as the water fills the pipes and flows toward the open heads.

11.0 Quality Inspection and Testing

11.1 Inspection and Test Plan (ITP) – Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

11.1.1 Piping

- Piping Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Welding Procedures, Qualifications, and Inspection
- Piping and Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Leak Testing and Pressure Testing
- Cleaning and Flushing
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

11.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

11.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

11.2 QA/QC Inspection & Punchlist – Prior to filling, leak testing, and pressure testing, QA/QC inspection & punchlist shall be performed on all installed piping and components to verify proper installation in accordance with construction drawings, project specifications, and industry standards. Perform a final P&ID line check to ensure the installation has been properly reinstated after the pressure testing, cleaning, flushing, etc.

11.3 Gross Leak Test - Gross leak test shall be performed via compressed Nitrogen or Air at pressure up to 5 psig, on metallic piping only. Walk the entire system to verify that no leaks are present and make repairs to any leaks that are identified.

11.4 Hydrostatic Pressure Test - All piping shall be hydronic pressure tested at a minimum of 200 psig (13.8 Bar) pressure for a minimum of 2 hours, as required by NFPA 13 Section 25.1.

Properly vent the system to remove air via all air vents, during the water filling process.

The actual hydronic pressure test procedure, including means, methods, testing pressures, and durations, shall be determined during the construction project based on project specific requirements. The procedure shall be reviewed and approved by Micron for use.

12.0 Startup Commissioning / Site Acceptance Test

12.1 Upon completion of the fire protection system installation, perform acceptance tests as required and outlined in NFPA 13 and as required by the AHJ.

12.2 The equipment/components listed below shall be tested at a minimum, but not limited to the following:

12.2.1 Fire Pumps

12.2.2 Check Valves / Backflow Preventers

12.2.3 Pressure Reducing Valves

12.2.4 Pressure Relief Valves

12.2.5 Pre-action Valves and Deluge Valves

12.2.6 All Control Valves

12.2.7 Flow Switches and Supervisory Switches

13.0 References

- NFPA 13, "Installation of Sprinkler Systems."
- NFPA 25, “Water Based Fire Protection Systems.”
- NFPA 318, “Standard for the Protection of Semiconductor Fabrication Facilities.”
- FM Datasheet DS 2-0, “Installation Guidelines for Automatic Sprinklers”
- FM Datasheet DS 7-7, "Semiconductors Fabrication Facilities”

14.0 Reference Specifications:

14.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 16**.

Table 16 – Specification Sections

Spec Section	Spec Contents
Section 15060	Pipe and Pipe Fittings
Section 15190	PVC and CPVC Piping Systems
Section 15101	Valves
Section 15120	Piping Specialties
Section 15121	Process piping Specialties
Section 15130	Indicating Devices
Section 15140	Supports and Anchors
Section 15190	Mechanical Identification
Section 15240	Mechanical Sound and Vibration Control
Section 15260	Piping Insulation
Section 15280	Equipment Insulation
Section 15330	Wet Pipe Sprinkler Sysetms
Section 15461	Process Exhaust System
Section 15515	Hydronic Specialties
Section 15540	Pumps
Section 15860	Exhaust Fans
Section 15861	FRP Exhaust Fans and Stacks
Section 15891	Metal HVAC Ductwork and Accessories
Section 15892	Metal Industrial Ductwork and Accessories
Section 15993	Air Systems Testing, Adjusting, and Balancing
Section 15170	Motors
Section 15171	Variable Frequency Drives

15.0 Document Control

15.1 Approval:

GLOBAL_FAC_GFTT_APPR
RISK_MGMT

15.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_ NOTIFY
FCG_F2_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10W_FAC_NOTIFY
FCG_F10N_FAC_NOTIFY
FCG_MTTW_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
MTB_FAC_ALL_NOTIFY

15.3 Final Viewer Access

All Micron Team Members at all Sites

15.4 Retention

Adherence to the requirements specified in
Global Facilities - Document Control Procedures

15.5 Review

Biennially (two years)

16.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	11/10/20